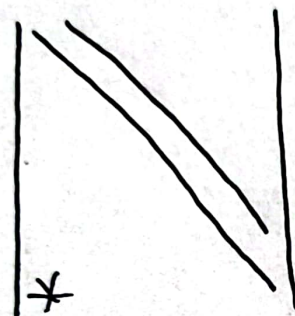


行列式类型总结

1. 双线型



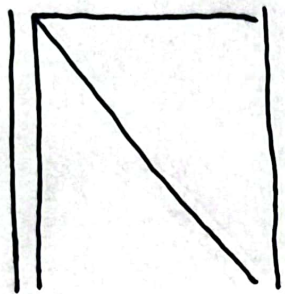
按“*”元素所在行/列展开

例: 计算 $D_n =$

$$\begin{vmatrix} a_1 & b_1 & & & \\ & a_2 & b_2 & & \\ & & & \ddots & \\ & & & & a_{n-1} & b_{n-1} \\ b_n & & & & & a_n \end{vmatrix}$$

$$D_n = \prod_{i=1}^n a_i + (-1)^{n+1} \prod_{j=1}^n b_j$$

2. 箭型

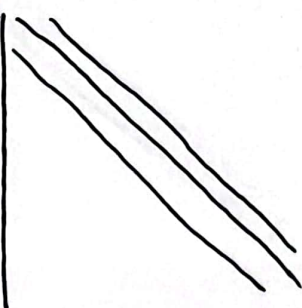


用对角线上的元素将第一行/列上的元素消去

例: 求 $D_{n+1} = \begin{vmatrix} a_0 & 1 & 1 & \cdots & 1 \\ 1 & a_1 & 0 & \cdots & 0 \\ 1 & 0 & a_2 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & 0 & 0 & \cdots & a_n \end{vmatrix} \quad (\forall i, a_i \neq 0)$

$$D_{n+1} = \begin{vmatrix} a_0 - \sum_{i=1}^n \frac{1}{a_i} & 1 & 1 & \cdots & 1 \\ C_1 - \frac{1}{a_1} C_2 & 0 & a_1 & 0 & \cdots & 0 \\ C_1 - \frac{1}{a_2} C_3 & 0 & 0 & a_2 & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ C_1 - \frac{1}{a_n} C_{n+1} & 0 & 0 & 0 & \cdots & a_n \end{vmatrix} = \left(\prod_{i=1}^n a_i \right) \cdot \left(a_0 - \sum_{i=1}^n \frac{1}{a_i} \right)$$

3. 三对角型



按第一行/列展开, 得到递推式 $D_n = \alpha D_{n-1} + \beta D_{n-2}$

考虑特征方程 $\lambda^2 - \alpha\lambda - \beta = 0$, 解得两根 x_1, x_2

则 $x_1 + x_2 = \alpha, x_1 x_2 = -\beta$, 从而 $D_n = (x_1 + x_2) D_{n-1} - x_1 x_2 D_{n-2}$

故: $D_n - x_1 D_{n-1} = x_2 (D_{n-1} - x_1 D_{n-2}) = \dots = x_2^{n-2} (D_2 - x_1 D_1)$ ①

$D_n - x_2 D_{n-1} = x_1 (D_{n-1} - x_2 D_{n-2}) = \dots = x_1^{n-2} (D_2 - x_2 D_1)$ ②

①. ② 加减消元即可

$$\text{例: } D_n = \begin{vmatrix} a+b & ab & & & \\ 1 & a+b & ab & & \\ & 1 & a+b & ab & \\ & & \ddots & \ddots & \ddots \\ & & & 1 & a+b \end{vmatrix} \quad (a \neq b)$$

$$D_n = (a+b)D_{n-1} - abD_{n-2}$$

$$\text{故 } D_n - aD_{n-1} = b(D_{n-1} - aD_{n-2}) = \dots = b^{n-2}(D_2 - aD_1)$$

$$D_n - bD_{n-1} = a(D_{n-1} - bD_{n-2}) = \dots = a^{n-2}(D_2 - bD_1)$$

$$\text{其中 } D_2 = \begin{vmatrix} a+b & ab \\ 1 & a+b \end{vmatrix} = a^2 + ab + b^2, \quad D_1 = a+b$$

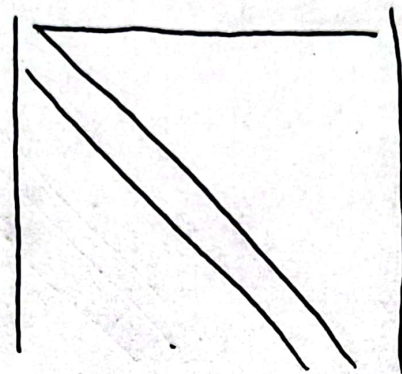
$$\text{故 } D_n = aD_{n-1} + b^n \quad ①$$

$$D_n = bD_{n-1} + a^n \quad ②$$

$$\text{①} \times b - \text{②} \times a \text{ 得 } (b-a)D_n = b^{n+1} - a^{n+1}$$

$$\text{所以 } D_n = \frac{b^{n+1} - a^{n+1}}{b-a}$$

4. Hessenberg 型



展开 / 倍加

例: $D_n = \begin{vmatrix} X & -1 & & & \\ & X & -1 & & \\ & & \ddots & \ddots & \\ & & & X & -1 \\ a_n & a_{n-1} & a_{n-2} & \cdots & a_2 & a_1 + X \end{vmatrix}$

5. 行列相等 将每列(行)加到第一列(行)上再消元

例: $D_n = \begin{vmatrix} X_{1-m} & X_2 & \cdots & X_n \\ X_1 & X_{2-m} & \cdots & X_n \\ \vdots & \vdots & & \vdots \\ X_1 & X_2 & \cdots & X_{n-m} \end{vmatrix}$

$$D_n \xrightarrow{\substack{C_1+C_2 \\ \vdots \\ C_1+C_n}} \begin{vmatrix} \sum_{i=1}^n X_i - m & X_2 & \cdots & X_n \\ \sum_{i=1}^n X_i - m & X_{2-m} & \cdots & X_n \\ \vdots & \vdots & & \vdots \\ \sum_{i=1}^n X_i - m & X_2 & \cdots & X_{n-m} \end{vmatrix} = \left(\sum_{i=1}^n X_i - m \right) \begin{vmatrix} 1 & X_2 & \cdots & X_n \\ 1 & X_{2-m} & \cdots & X_n \\ \vdots & \vdots & & \vdots \\ 1 & X_2 & \cdots & X_{n-m} \end{vmatrix}$$

$$\xrightarrow{\substack{C_2-X_2C_1 \\ \vdots \\ C_n-X_nC_1}} \begin{vmatrix} 1 & 0 & \cdots & 0 \\ 1 & -m & \cdots & 0 \\ \vdots & \vdots & & \vdots \\ 1 & 0 & \cdots & -m \end{vmatrix} \cdot \left(\sum_{i=1}^n X_i - m \right) = (-m)^{n-1} \cdot \left(\sum_{i=1}^n X_i - m \right)$$

6. 滚动型 倍加

例: $D_n =$

$$\begin{vmatrix} 1 & a & a^2 & \cdots & a^{n-2} & a^{n-1} \\ a^{n-1} & 1 & a & \cdots & a^{n-3} & a^{n-2} \\ a^{n-2} & a^{n-1} & 1 & \cdots & a^{n-4} & a^{n-3} \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ a^2 & a^3 & a^4 & \cdots & 1 & a \\ a & a^2 & a^3 & \cdots & a^{n-1} & 1 \end{vmatrix}$$

$D_n \xrightarrow{\substack{R_1 - aR_2 \\ R_2 - aR_3 \\ \vdots \\ R_{n-1} - aR_n}}$

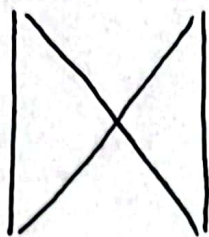
$$\begin{vmatrix} 1-a^n & 0 & 0 & \cdots & 0 & 0 \\ 0 & 1-a^n & 0 & \cdots & 0 & 0 \\ 0 & 0 & 1-a^n & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 1-a^n & 0 \\ a & a^2 & a^3 & \cdots & a^{n-1} & 1 \end{vmatrix} = (1-a^n)^{n-1}$$

7. Vandermonde 行列式

例: $D_n =$

$$\begin{vmatrix} 1 & 1 & \cdots & 1 \\ 1+\sin\varphi_1 & 1+\sin\varphi_2 & \cdots & 1+\sin\varphi_n \\ \sin\varphi_1+\sin^2\varphi_1 & \sin\varphi_2+\sin^2\varphi_2 & \cdots & \sin\varphi_n+\sin^2\varphi_n \\ \vdots & \vdots & \ddots & \vdots \\ \sin^{n-2}\varphi_1+\sin^{n-1}\varphi_1 & \sin^{n-2}\varphi_2+\sin^{n-1}\varphi_2 & \cdots & \sin^{n-2}\varphi_n+\sin^{n-1}\varphi_n \end{vmatrix}$$

8. Laplace 定理



$$\text{例: } D_{2n} = \begin{vmatrix} n & & & & n+1 \\ & n-1 & & & \\ & & \ddots & & \\ & & & 1 & 2 \\ & & & & \ddots & n \\ n+1 & & & 3 & 4 & \\ & & & & & \ddots & n+2 \\ & & & & & & n+3 \end{vmatrix}$$

$$D_{2n} = \begin{vmatrix} n & n+1 \\ n+2 & n+3 \end{vmatrix} D_{2n-2} = -2 D_{2n-2} = \dots = (-2)^{n-1} D_2$$

$$D_2 = \begin{vmatrix} 1 & 2 \\ 3 & 4 \end{vmatrix} = -2$$

$$\text{故 } D_{2n} = (-2)^n$$

9. 拆分法

$$\text{例: } D_n = \begin{vmatrix} x_1 & a & \cdots & a \\ a & x_2 & \cdots & a \\ \vdots & \vdots & & \vdots \\ a & a & \cdots & x_n \end{vmatrix}$$

$$D_n = \begin{vmatrix} x_1 & a & \cdots & a \\ a & x_2 & \cdots & a \\ \vdots & \vdots & & \vdots \\ a & a & \cdots & a \end{vmatrix} + \begin{vmatrix} x_1 & a & \cdots & 0 \\ a & x_2 & \cdots & 0 \\ \vdots & \vdots & & \vdots \\ a & a & \cdots & x_n a \end{vmatrix}$$

$$= a \prod_{i=1}^{n-1} (x_i - a) + (x_n - a) D_{n-1}$$

$$\text{变型: } D_n = \begin{vmatrix} x_1 & a & \cdots & a \\ b & x_2 & \cdots & a \\ \vdots & \vdots & & \vdots \\ b & b & \cdots & x_n \end{vmatrix}$$

$$D_n = \begin{vmatrix} d & b & b & \cdots & b \\ c & x & a & \cdots & a \\ c & a & x & \cdots & a \\ \vdots & \vdots & \vdots & & \vdots \\ c & a & a & \cdots & x \end{vmatrix}$$

10. 加边法

例: $D_n = \begin{vmatrix} 1+x_1^2 & x_2x_1 & \cdots & x_nx_1 \\ x_1x_2 & 1+x_2^2 & \cdots & x_nx_2 \\ \vdots & \vdots & \ddots & \vdots \\ x_1x_n & x_2x_n & \cdots & 1+x_n^2 \end{vmatrix}$

$$D_n = \begin{vmatrix} 1 & x_1 & x_2 & \cdots & x_n \\ 0 & 1+x_1^2 & x_2x_1 & \cdots & x_nx_1 \\ 0 & x_1x_2 & 1+x_2^2 & \cdots & x_nx_2 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & x_1x_n & x_2x_n & \cdots & 1+x_n^2 \end{vmatrix} \xrightarrow{\substack{R_2 - x_1R_1 \\ R_3 - x_2R_1 \\ \vdots \\ R_{n+1} - x_nR_1}} \begin{vmatrix} 1 & x_1 & x_2 & \cdots & x_n \\ -x_1 & 1 & 0 & \cdots & 0 \\ -x_2 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ -x_n & 0 & 0 & \cdots & 1 \end{vmatrix}$$